Counter Galois Onion (CGO)

A New Onion Encryption Scheme for Tor

Jean Paul Degabriele, Alessando Melloni, Jean-Pierre Münch, Martijn Stam

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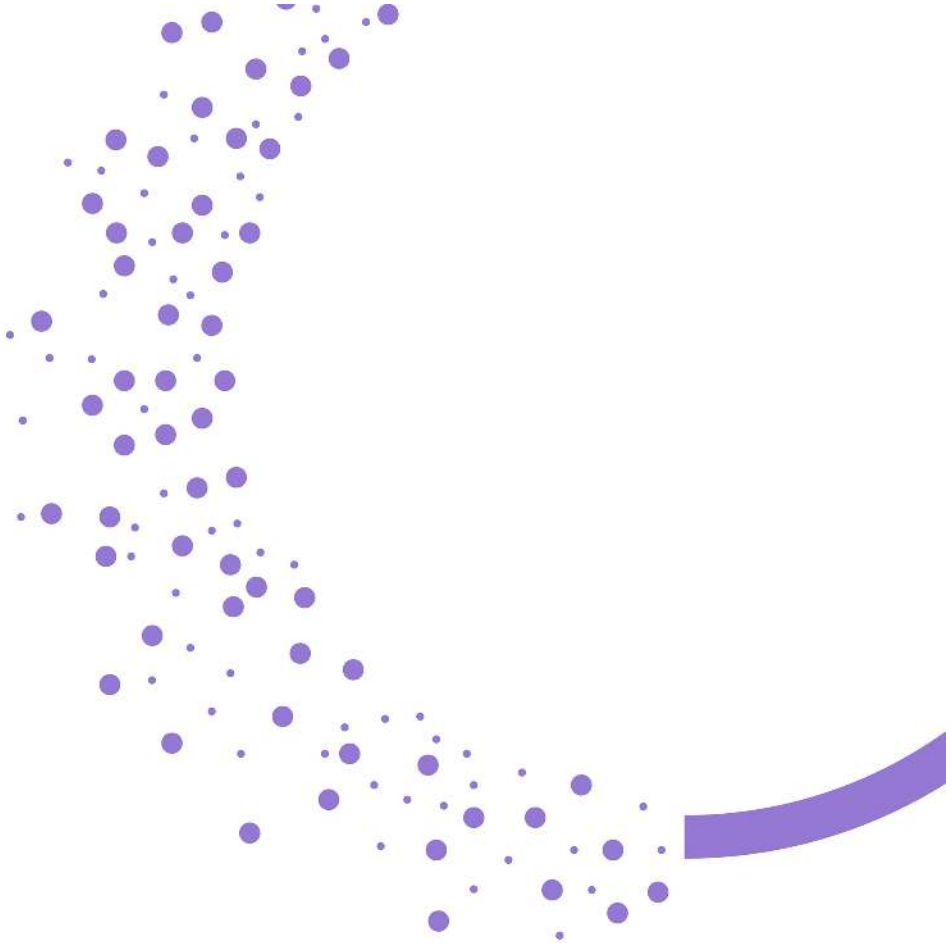
Tor's Onion Encryption and its Limitations



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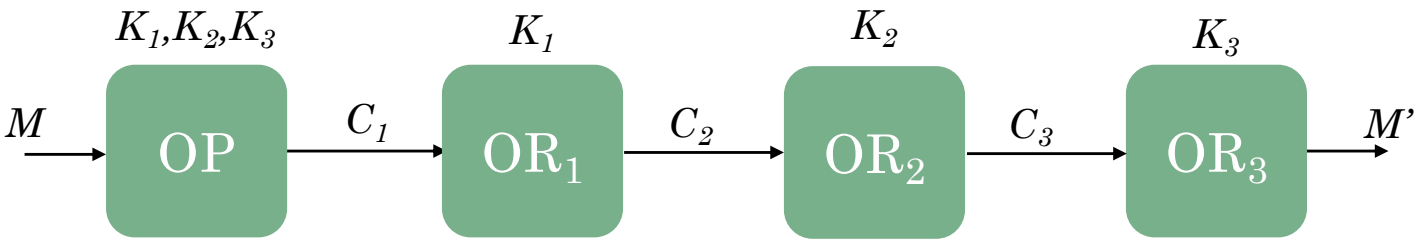

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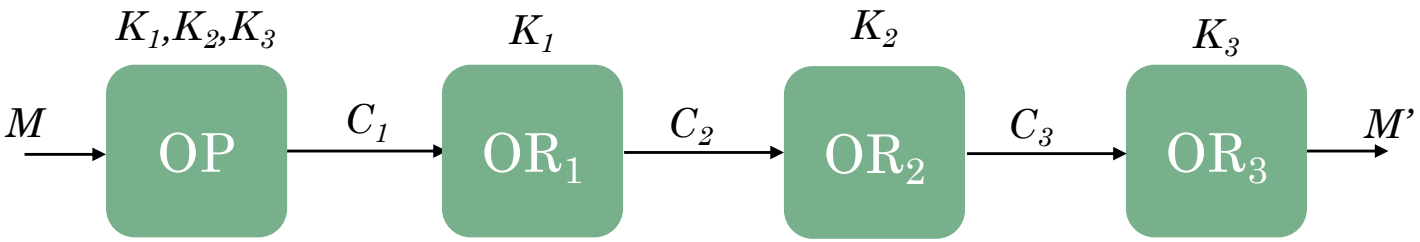
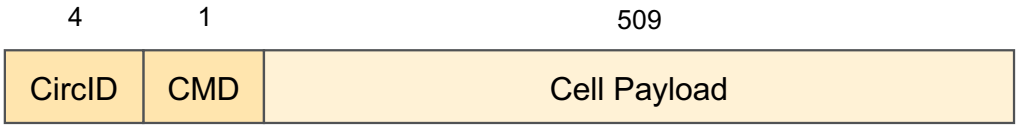


- In Tor, data flows through **circuits** consisting of an **Onion Proxy (OP)** and a sequence of **Onion Routers (OR)**.
- Circuits are **analogues of symmetric channels**, where the Onion Proxy shares a symmetric key with each Onion Router.
- Each Onion Router **"peels off"** a layer of encryption, **decorrelating incoming ciphertexts** from **outgoing ciphertexts**.

Relay Cell Format and Processing [DMS04]

- Cells are 514 bytes (v4+)



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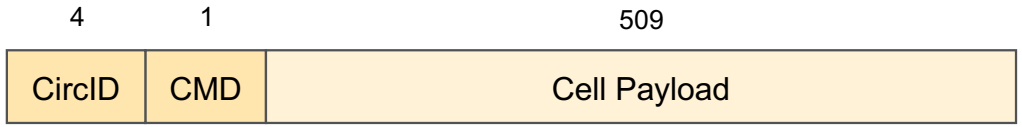

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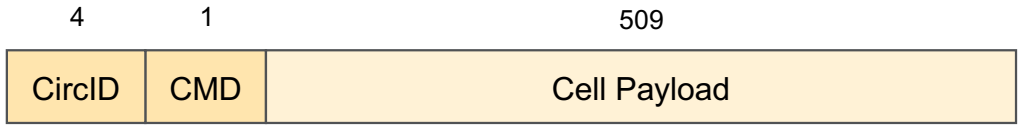

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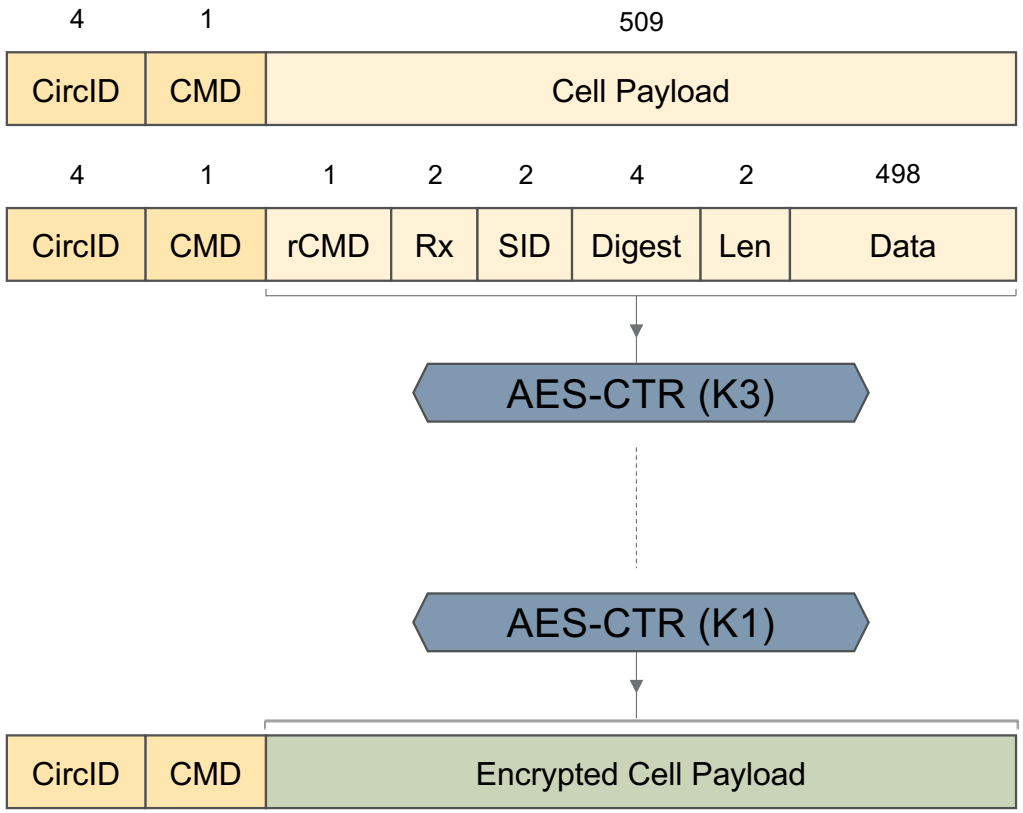

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- **Bits** are flipped in a ciphertext at $\mathbf{OR_1}$ and reversed at $\mathbf{OR_3}$ to **test** whether **decryption succeeds** and ciphertexts belong to the **same circuit**.

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Historical Timeline of Evolving Solutions

[2012] Tor Proposal 202

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- Specifically, it called for Onion Encryption that:

  1. Protects against **Tagging Attacks**,
  2. Is **non-expanding**, so as not to leak circuit size and relative position,
  3. Offers **Stronger end-to-end integrity** protection,
  4. Does not use **MAC-then-Encrypt**.

Proposed Solutions

- In **Prop 261(2016)** Matthewson proposed a new Onion Encryption scheme where each encryption layer consists of a **Wide-Input Tweakable Cipher.**

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- In **[ADL17]** GCM-RUP is also proposed for **use in Tor**, claiming that a **WITC is overkill**, an OE scheme is described but **without any formal analysis**.

Onion Encryption from GCM-RUP

- In [**ADL17**] the Onion Encryption scheme was **underspecified**, **no integrity**, **no leaky pipes**, **no ciphertext chaining**, and **not forward secure**.

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- **[Prop 295-Mar 2019]** Heavily revised scheme, made stateful by **feeding back GHASH outputs** (invalidating proofs) and **extending GCM-RUP to full WITC (SPRP),** forfeiting the performance gains originally sought!

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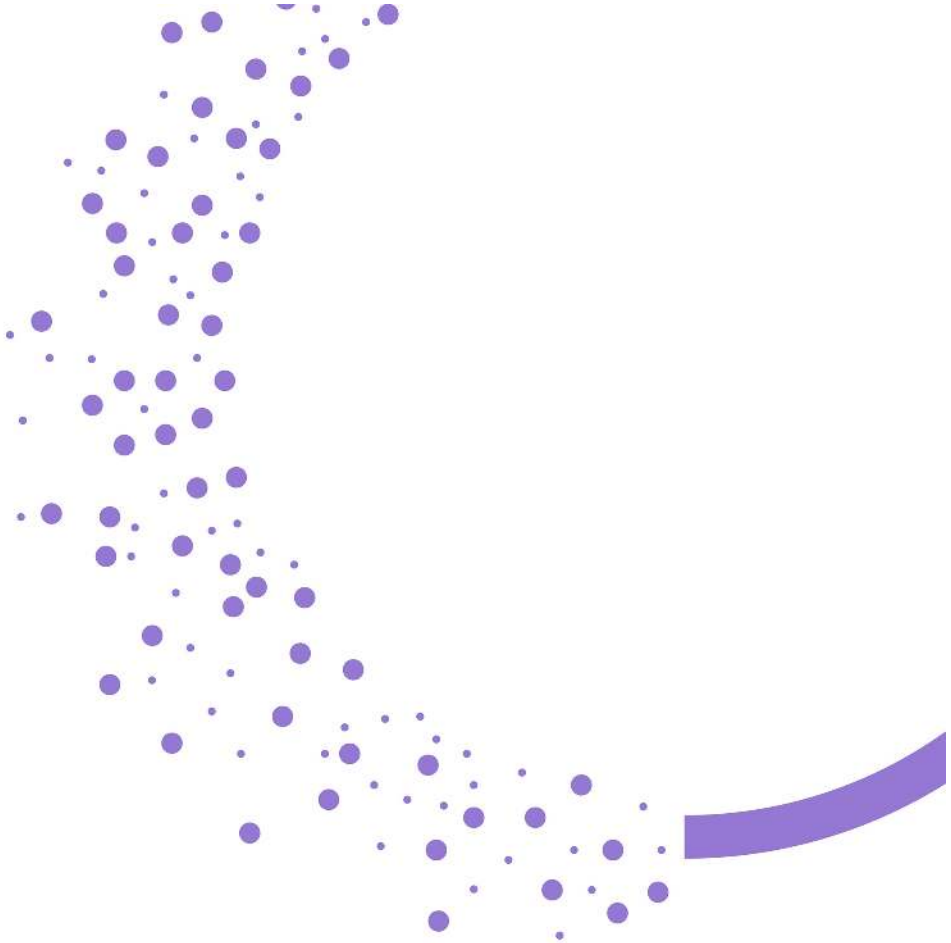
The Design of CGO



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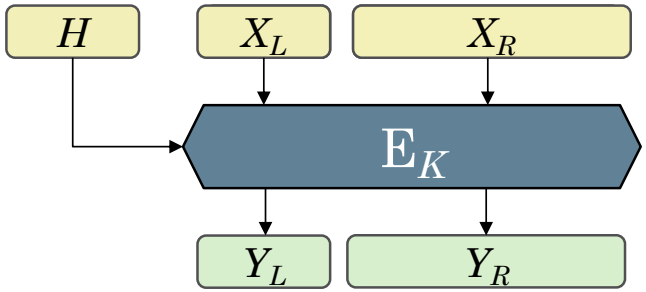
- A new formal treatment of OE extending prior ones [DegSta18, RogZha18] with **Leaky Pipes**, **Associated Data**, **Forward Security**.
- We build CGO in a modular fashion from any **Updateable Rugged PRP** and **prove it secure** in our new model.

Main Building Block: Rugged PRPs



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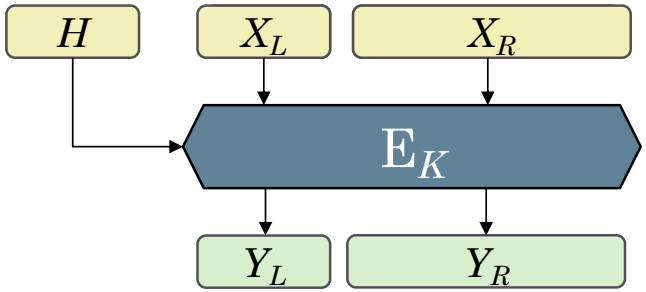

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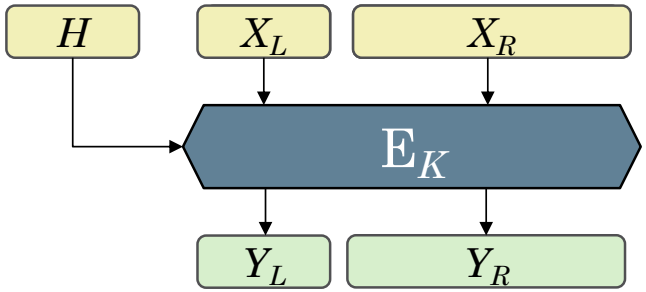

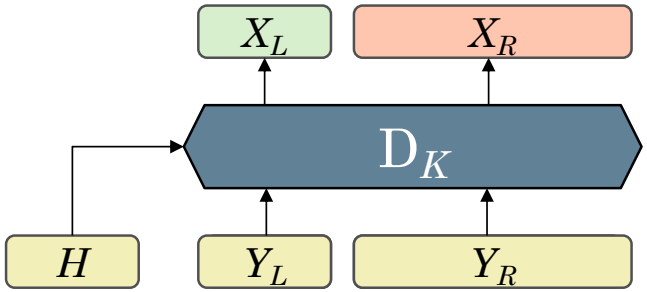

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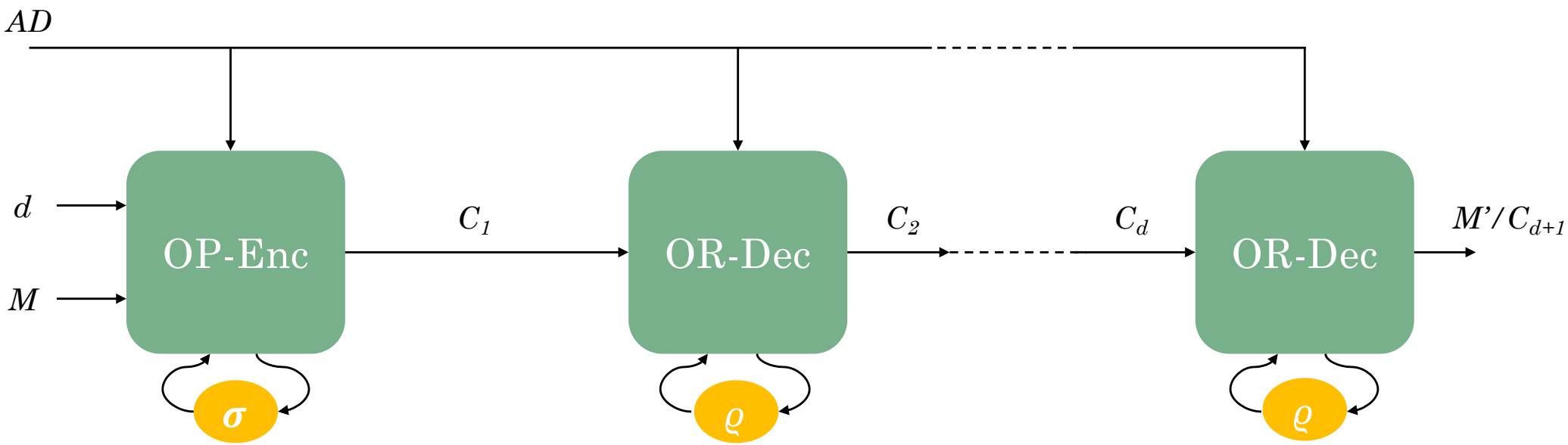

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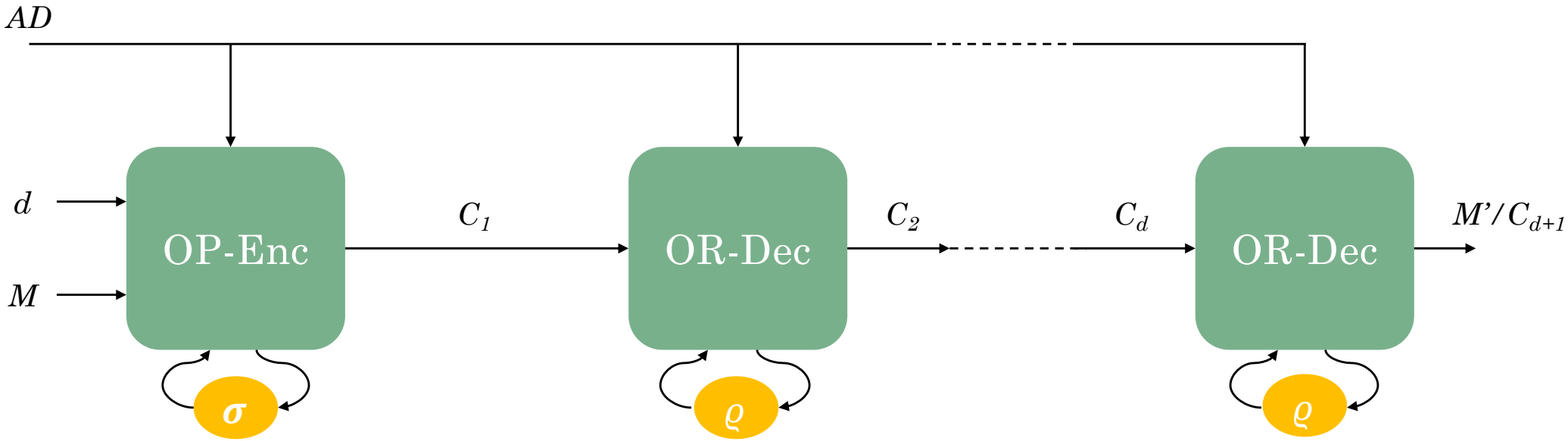

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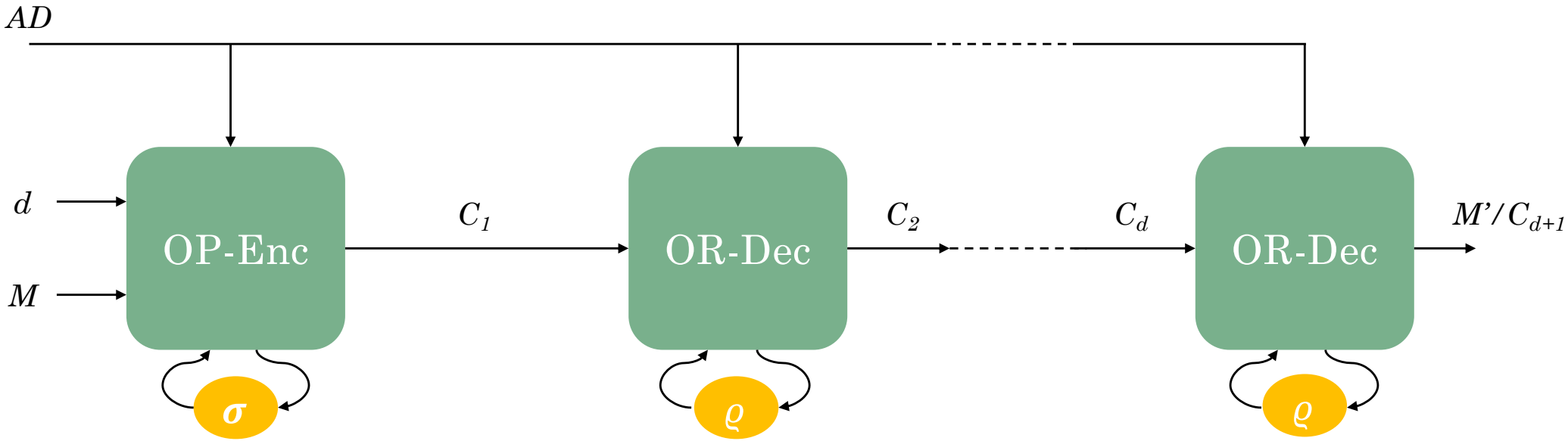

- **Note:** decryption cannot detect invalid ciphertexts – **does not return** ⊥.
- Hence the need to **chain ciphertexts** - if a **ciphertext is tampered** subsequent ciphertexts **decrypt to garbage**.

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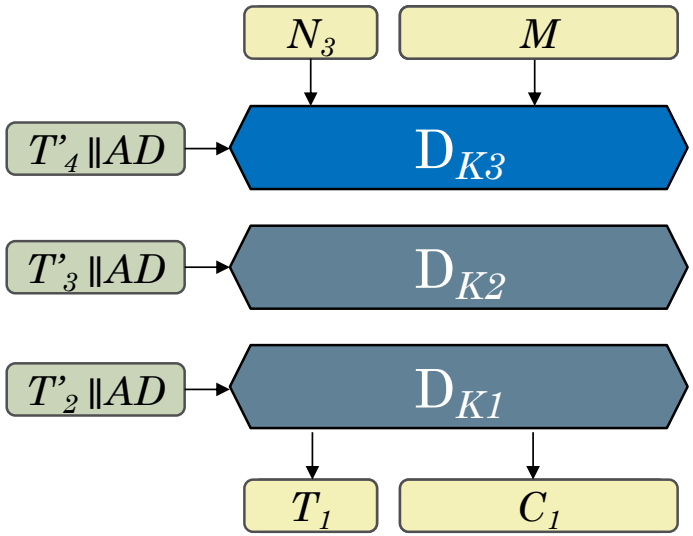

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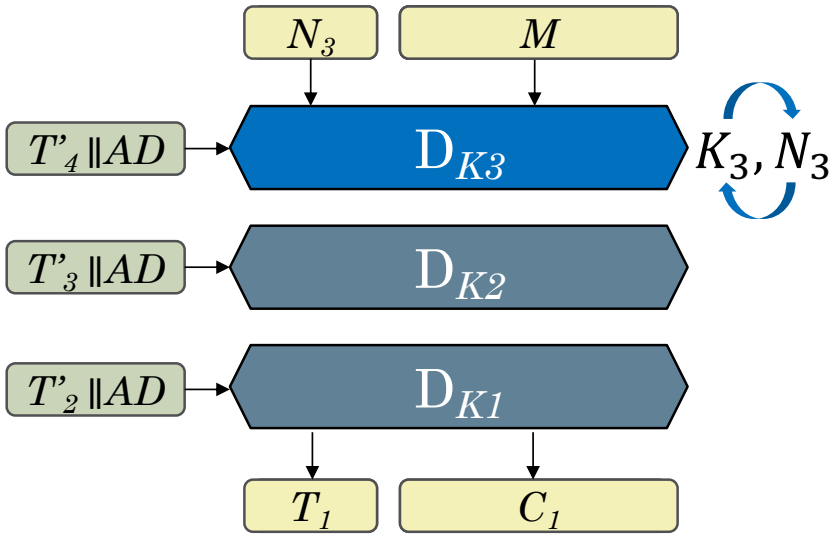

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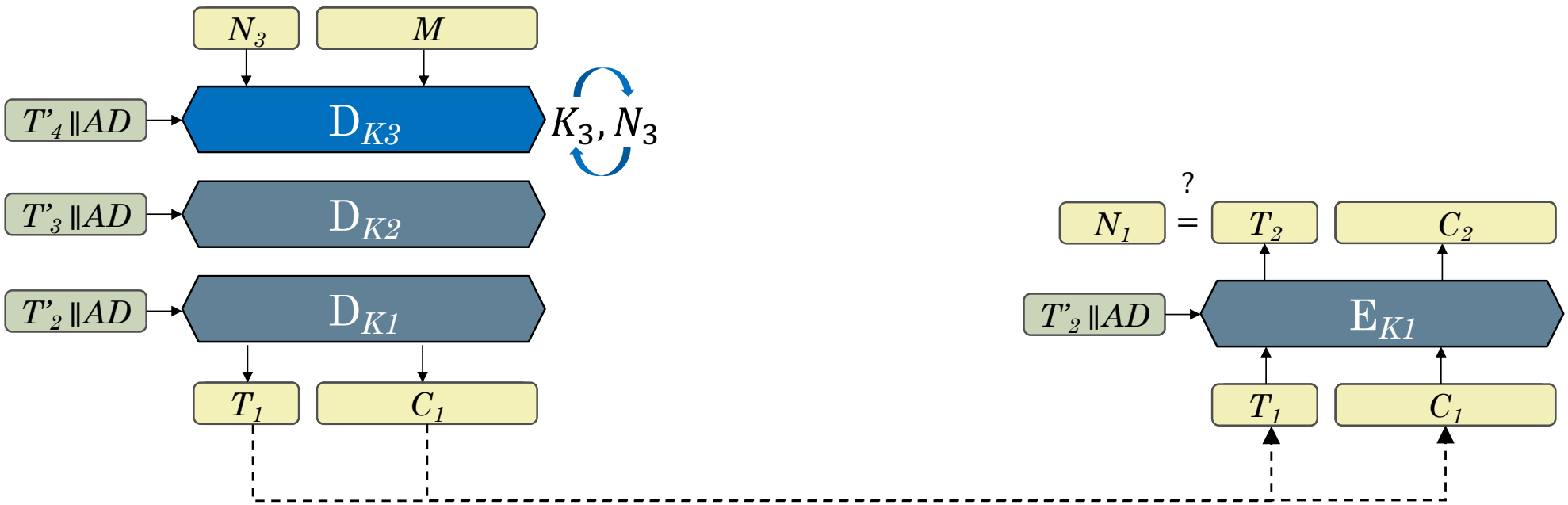

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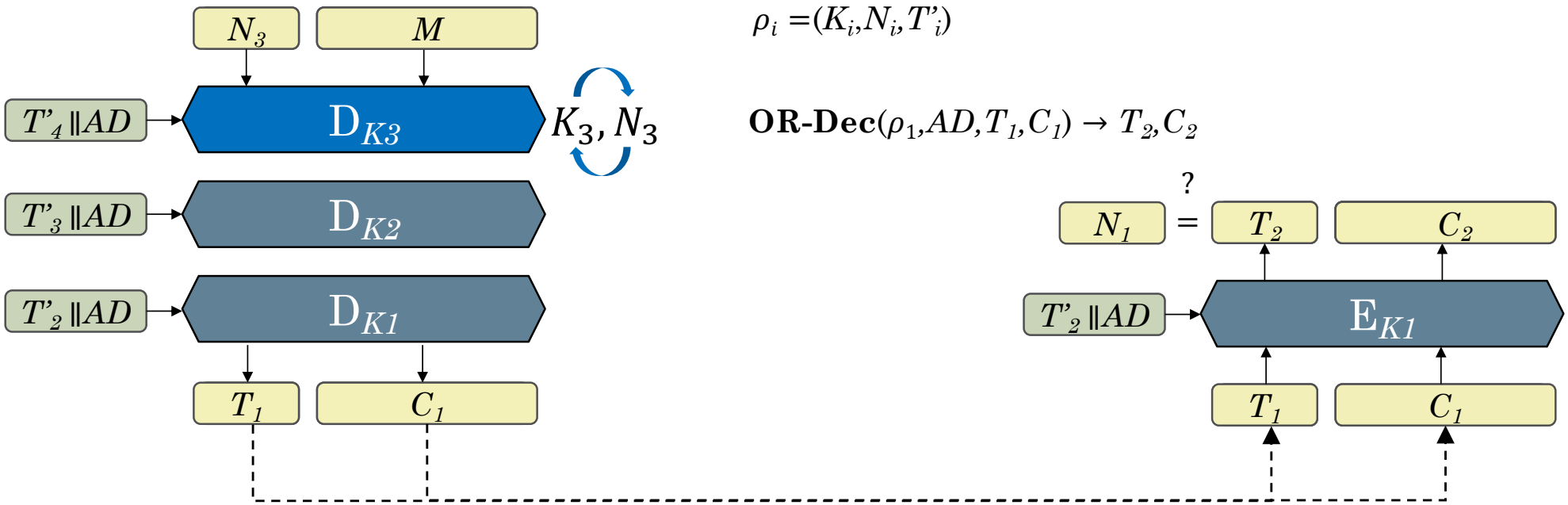

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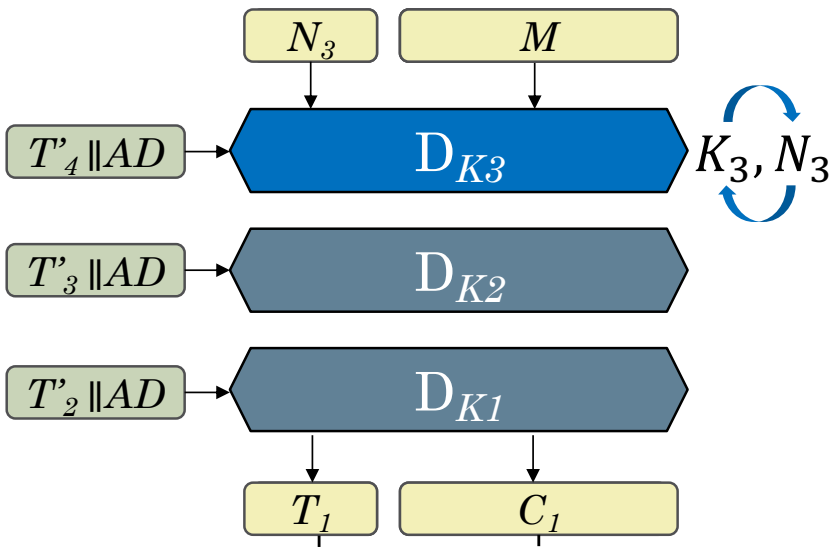

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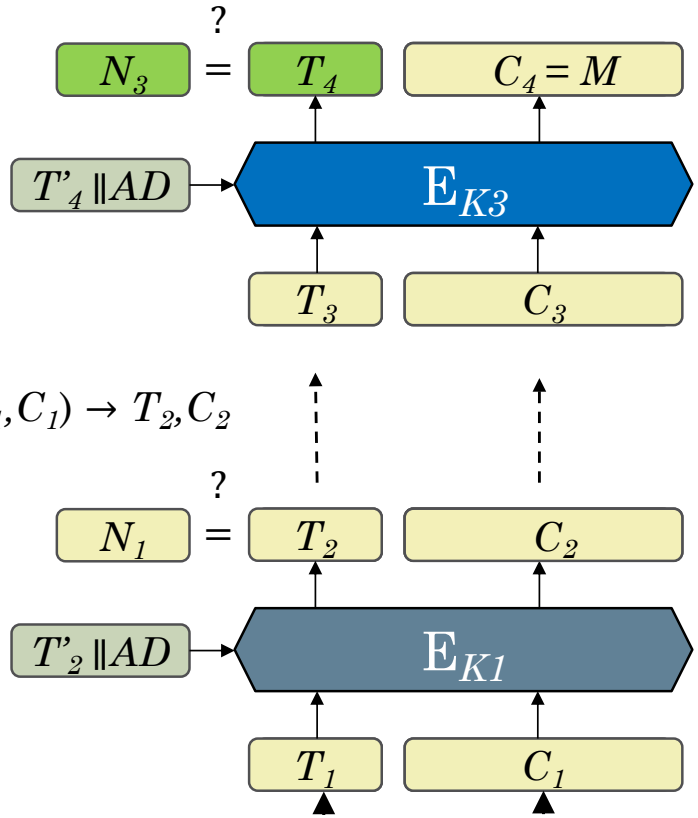

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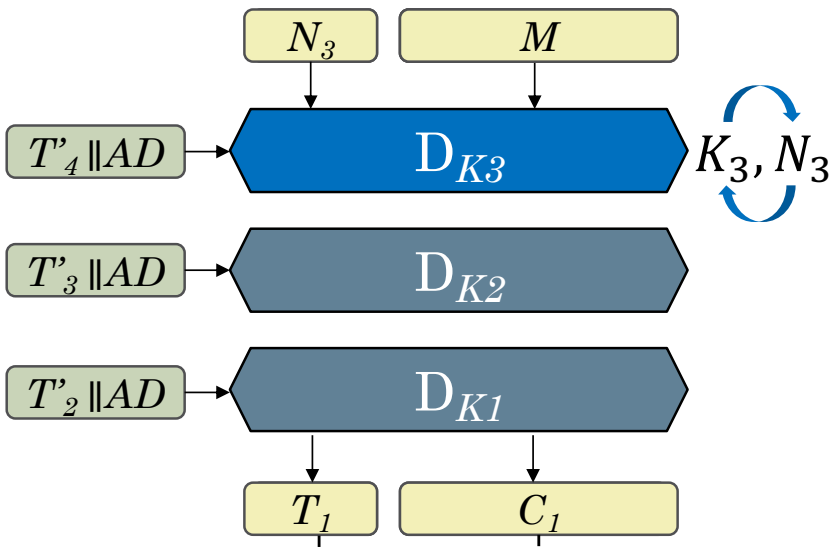

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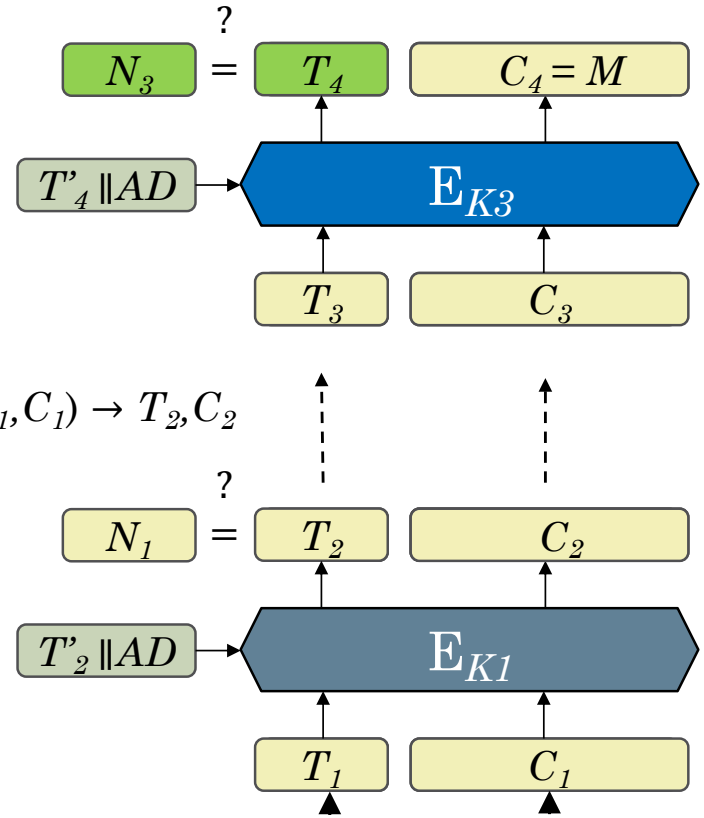

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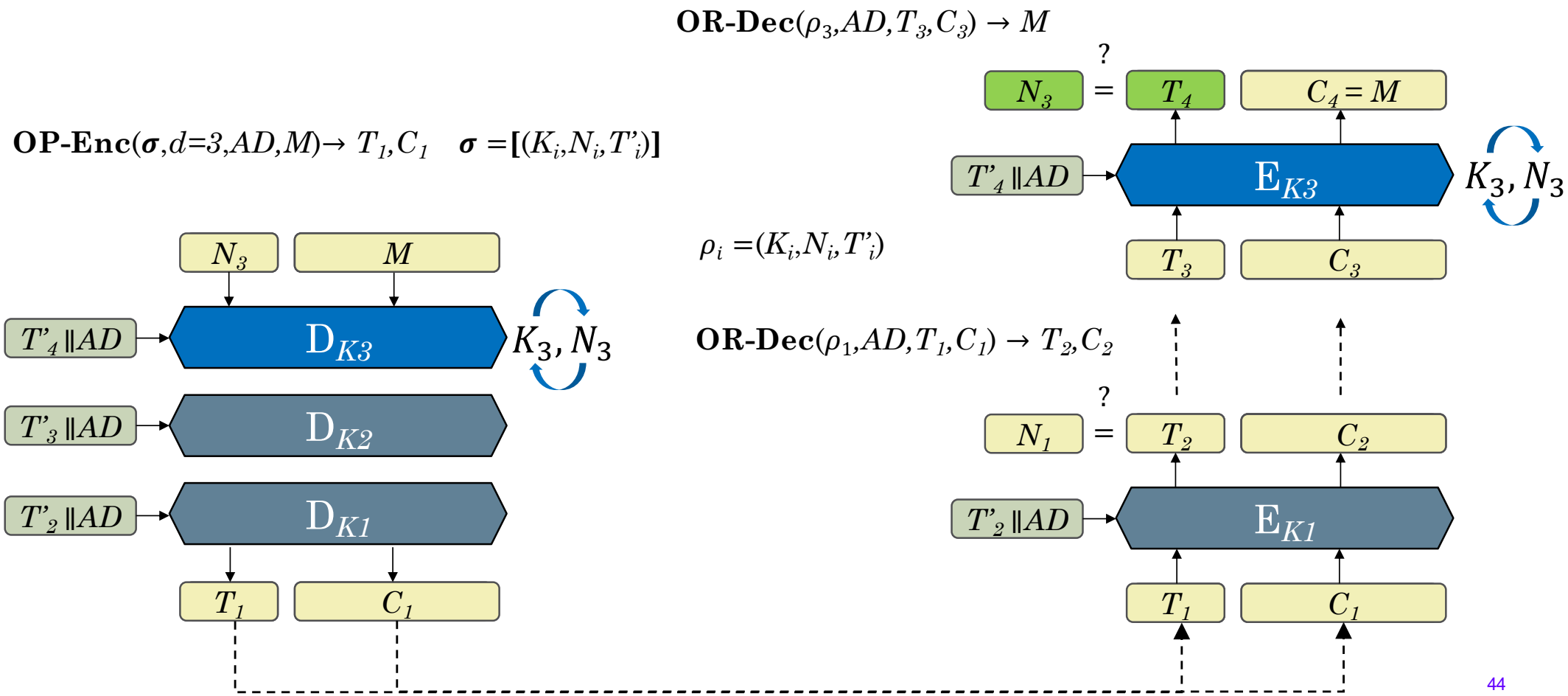

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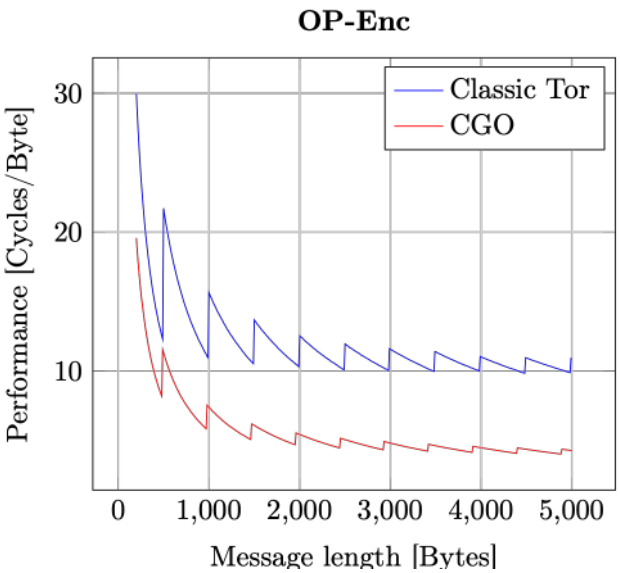
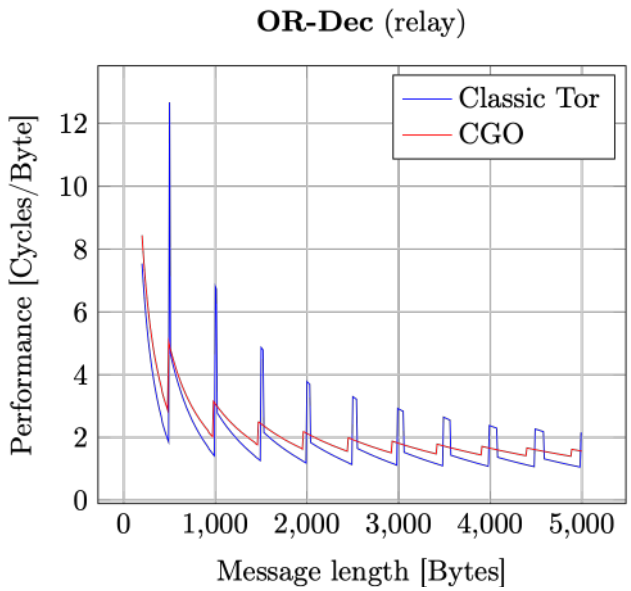
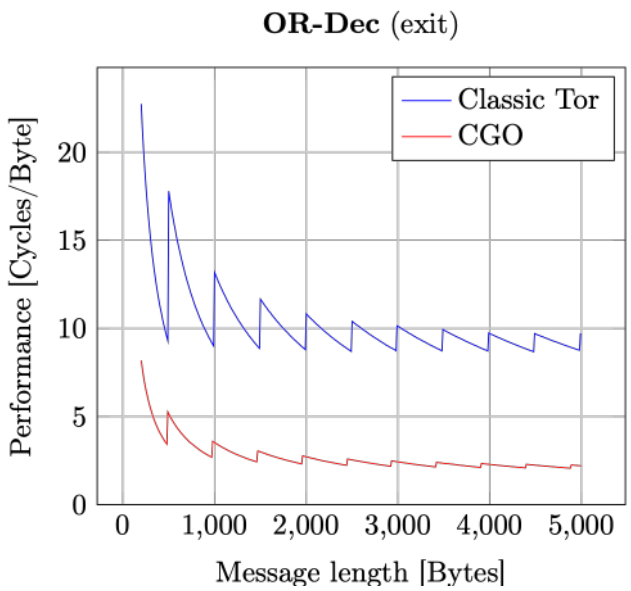
- **Associated Data**, prevents tagging in the headers (**CMU 2014 Attack**).

CGO Performance







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- CGO is currently being integrated in **Arti**—Tor's upcoming Rust implementation.
- https://eprint.iacr.org/2025/583
- https://eprint.iacr.org/2025/2017